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NUMERICAL METHODS FOR THE JIN-XIN MODEL

IMEX SCHEMES AND RELAXNN FRAMEWORKS

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Kinetic Equations: Classical Foundations and Modern Computational Perspectives Examination
(July 23, 2026)

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Introduction & Mathematical Background

Implicit-Explicit Methods & Relax Neural Networks

Implicit Explicit Neural Network (Proposal)

REFERENCES

- [1] S. Jin and Z. Xin, “**The relaxation schemes for systems of conservation laws in arbitrary space dimensions,**” *Communications on Pure and Applied Mathematics*, 1995.
- [2] U. M. Ascher, S. J. Ruuth, and R. J. Spiteri, “**Implicit-explicit runge-kutta methods for time-dependent partial differential equations,**” *Applied Numerical Mathematics*, 1997.
- [3] S. Boscarino, L. Pareschi, and G. Russo, *Implicit-Explicit Methods for Evolutionary Partial Differential Equations*. Society for Industrial and Applied Mathematics, 2024.
- [4] N. Zhou and Z. Ma, ***Capturing shock waves by relaxation neural networks***, 2024.

INTRODUCTION & MATHEMATICAL BACKGROUND

THE TARGET & JIN-XIN RELAXATION APPROXIMATION

Target: Viscous/Inviscid Burgers' Equation

$$\partial_t u + \partial_x (f(u)) = 0 \quad \xrightarrow{\text{viscous reg.}} \quad \partial_t u + u \partial_x u = \nu \partial_{xx} u$$

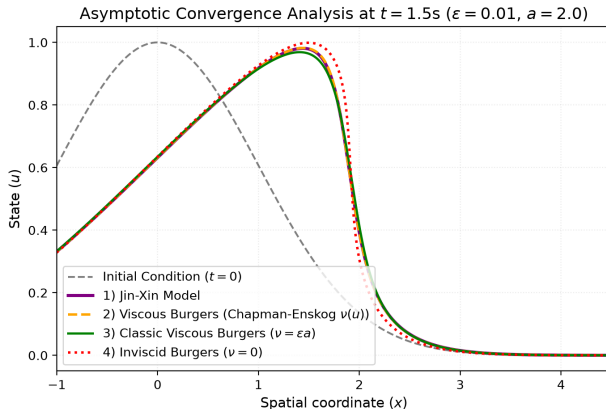
Linear Hyperbolic System with Stiff Relaxation (Jin-Xin)

$$\begin{cases} \partial_t u + \partial_x v = 0 \\ \partial_t v + a \partial_x u = -\frac{1}{\varepsilon} (v - f(u)) \end{cases}$$

Key Properties

- **Characteristic Speeds:** Fixed linear velocities $\pm\sqrt{a}$ (Riemann Invariants).
- **Sub-characteristic Condition:** $a > (f'(u))^2$ guarantees linear stability.

ASYMPTOTIC EQUIVALENCE & FOUNDATIONAL MOTIVATIONS



Chapman-Enskog ($\varepsilon \rightarrow 0$) :
$$\partial_t u + \partial_x f(u) = \partial_x \left[\varepsilon \left(a - (f'(u))^2 \right) \partial_x u \right] + \mathcal{O}(\varepsilon^2)$$

IMPLICIT-EXPLICIT METHODS & RELAX NEURAL NETWORKS

$$\begin{cases} \partial_t u + \nabla_x \cdot \mathbf{v} = 0 \\ \partial_t \mathbf{v} + a \nabla_x u = -\frac{1}{\varepsilon} (\mathbf{v} - \mathbf{f}(u)) \end{cases}$$

- **Classical Bottleneck:** Finite Volume & IMEX scale poorly in high-dimensional settings (*Curse of Dimensionality*) and are bounded by mesh CFL constraints.
- **SciML Paradigm Shift:** Neural Operators act as **mesh-free, continuous maps**. Inference is $\mathcal{O}(1)$, invariant to spatial resolution or parameters (ε, a) .

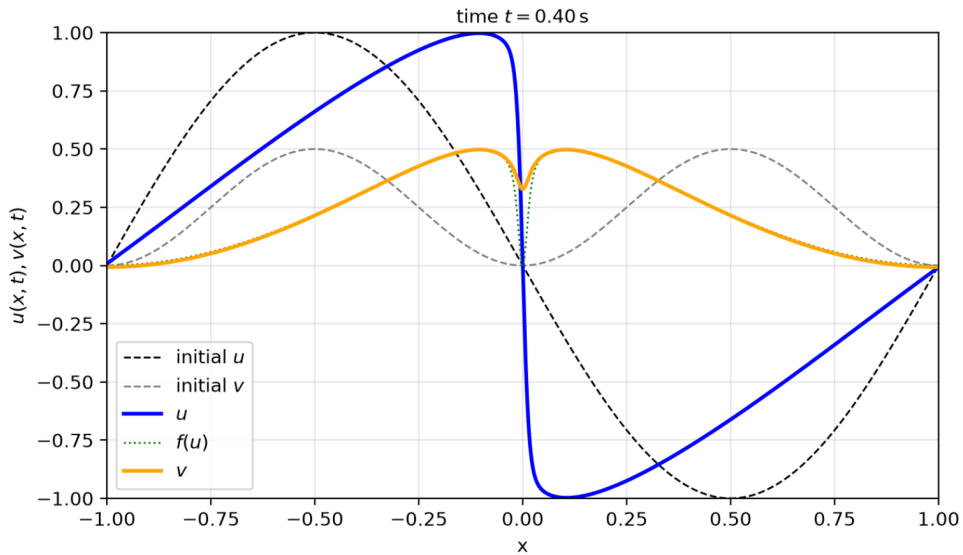
Operator Splitting for Multi-Scale Systems ($\varepsilon \ll 1$)

- **Explicit Step (Convection):** Resolve linear advection via Rusanov flux:

$$u^{n+1} = u^n - \Delta t \nabla_x \cdot \mathbf{v}^n$$

- **Implicit Step (Relaxation):** Treat stiff algebraic source implicitly:

$$\mathbf{v}^{n+1} = \frac{\varepsilon \mathbf{v}^n - \varepsilon \Delta t a \nabla_x u^n + \Delta t \mathbf{f}(u^{n+1})}{\varepsilon + \Delta t}$$



- **Chapman-Enskog Decoupling:** Micro-fluctuation $\mathbf{w}$ ($\mathbf{v} = \mathbf{f}(u) + \varepsilon \mathbf{w}$):

$$\begin{cases} \partial_t u + \nabla_x \cdot \mathbf{f}(u) = -\varepsilon \nabla_x \cdot \mathbf{w} \\ -\partial_t \mathbf{f}(u) - a \nabla_x u = \mathbf{w} + \varepsilon \partial_t \mathbf{w} \end{cases}$$

- **Asymptotic Preserving (AP) Loss** ($\varepsilon \rightarrow 0$): Forces equilibrium manifold convergence:

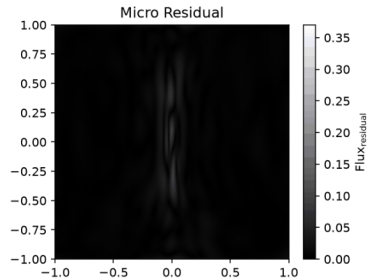
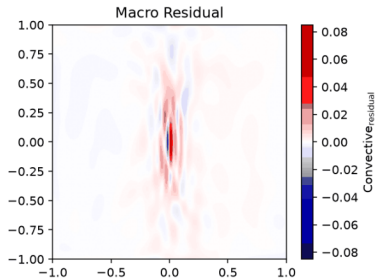
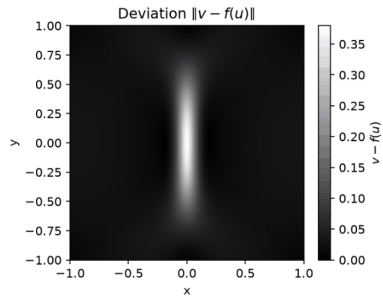
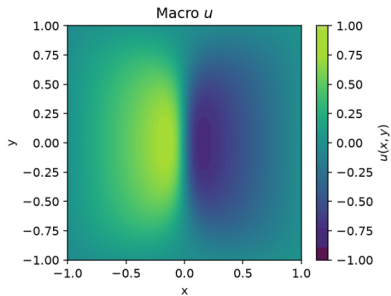
$$\mathcal{L}_0(\tilde{u}, \tilde{\mathbf{w}}) = \text{MSE}(\partial_t \tilde{u} + \nabla_x \cdot \mathbf{f}(\tilde{u}), 0) + \text{MSE}(-\partial_t \mathbf{f}(\tilde{u}) - a \nabla_x \tilde{u}, \tilde{\mathbf{w}})$$

Remark: Decoupling acts as an ε -shield, avoiding high-frequency oscillations and reducing training time from $\sim 300\text{k}$ to standard budgets.

- **Network Topology:** Shared deep **SiLU Encoder** branching into decoupled heads for macro ($\tilde{u}$) and micro ($\tilde{w}$) fields.
- **Geometric Shock Shaping:** Final Tanh layer mimics analytical viscous shock:

$$u(\mathbf{x}, t) = \frac{u_L + u_R}{2} - \frac{u_L - u_R}{2} \tanh \left(\frac{(u_L - u_R)(\mathbf{x} - st)}{4\nu} \right)$$

- **Loss Formulation:** Combined MSE (boundaries/IC) and MAE $\mathcal{L}_1$ (residuals for sharp shock capturing) + L_1/L_2 regularizations.



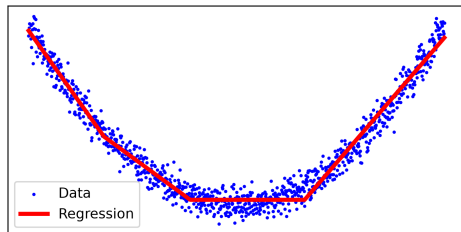
IMPLICIT EXPLICIT NEURAL NETWORK (PROPOSAL)

MOTIVATION AND THE CORE IDEA

The Challenge: Sub-grid information loss on coarse grids. Classical stencils (Rusanov) trade off numerical diffusion vs unphysical oscillations.

Objective: Optimize *only* the discrete spatial operator via data, embedded in a conservative IMEX framework.

- Circumvents black-box temporal tracking.
- Operates as a physics-bounded spatial regression function.



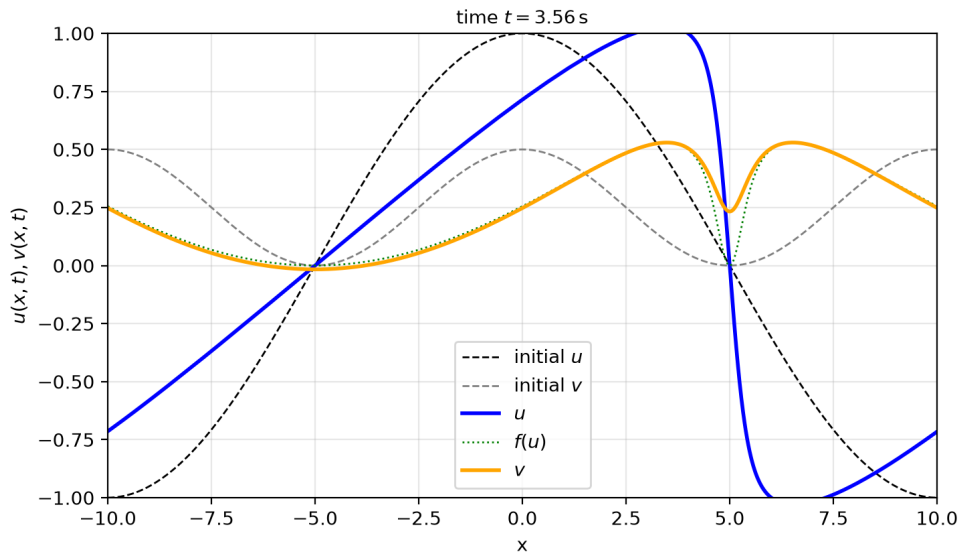
Stability Traps

- **Manifold Attraction:** Overfitting to physical diffusion instead of fluxes.
- **Central Bias:** FCNs default to non-dissipative symmetric stencils, triggering explosions.

Constrained Optimization Framework

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{interp}} + \lambda_1 \mathcal{L}_{\text{additivity}} + \lambda_2 \mathcal{L}_{\text{stability}}$$

- $\mathcal{L}_{\text{interp}}$: Local error vs reference gradients.
- $\mathcal{L}_{\text{additivity}}$: Semi-group coherence ($\Delta t/2 \times 2 = \Delta t$).
- $\mathcal{L}_{\text{stability}}$: Spectral radius control via Fourier Analysis.



Remark

The data-driven spatial operator is a structurally bounded optimization conditioned by training distribution.

Core Research Directions

- **Function Space Adaptation:** Family of profiles identified during training and ReLU activation mapping.
- **Shock Specialization:** Can non-linearities act as dynamic, data-driven flux limiters without manual tuning?



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*ADAMUS Project, funded by the Italian Ministry of University and Research (MUR) under
Directorial Decree No. 1236 of Aug 1, 2023 - Italian Science Fund - Call FIS2, CUP
F53C25000210001*



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